A REPORT

ON

(Traffic Signal Simulation System with Priority Vehicle Handling)

By

Name	Registration Number
1. Vedashree Patil	AP24110011507
2. Spoorthi Sri	AP24110011479
3. Bhima Tarini	AP24110011469
4. Akshara Putchakayala	AP24110011599

Prepared in the partial fulfillment of the

Project Based Learning of Course CSE 206 – Coding Skills II

SRM UNIVERSITY, AP

APRIL 2026

ABSTRACT

Traffic management is an essential component of modern urban infrastructure, where efficient handling of vehicles at signals is crucial to avoid congestion and delays. This project presents a **Traffic Signal Simulation System using the Queue Data Structure**, implemented in the C programming language.

The system follows the **First-In-First-Out (FIFO)** principle to manage vehicles waiting at a traffic signal. To enhance real-world applicability, the project incorporates a **priority-based mechanism**, where emergency vehicles such as ambulances, fire trucks, and police vehicles are given higher priority over normal vehicles.

Two separate queues are maintained: a **priority queue** for emergency vehicles and a **normal queue** for regular vehicles. When the traffic signal turns green, vehicles from the priority queue are processed first, followed by vehicles from the normal queue.

This simulation demonstrates the practical use of data structures in real-life scenarios and helps in understanding how queue operations can be applied in traffic management systems. The project also highlights the importance of prioritization in emergency situations, ensuring faster response and improved efficiency.

INDEX

S.No	Content	Page No.
1	Introduction	04
2	Problem Statement	05
3	Objectives	06
4	Data Structure Used	07
5	Algorithm	09
6	System WorkFlow	11
7	Implementation	12
8	Advantages and Limitations	13
9	Conclusion	14

Introduction:

Traffic management is one of the most important challenges in modern cities due to the increasing number of vehicles and limited road infrastructure. At traffic signals, vehicles are generally managed in an orderly manner to avoid congestion and ensure smooth flow of traffic. However, in real-world scenarios, emergency vehicles such as ambulances, fire trucks, and police vehicles require immediate passage without delay.

To model and understand this system, this project implements a **Traffic Signal Simulation using the Queue Data Structure in C programming**. The queue follows the **First-In-First-Out (FIFO)** principle, where the vehicle that arrives first is served first. In addition to this basic concept, the project introduces a **priority-based mechanism** to handle emergency vehicles more efficiently.

Two separate queues are used in the system: a **priority queue** for emergency vehicles and a **normal queue** for regular vehicles. This ensures that emergency vehicles are processed before normal traffic, simulating real-world traffic control systems.

The main aim of this project is to demonstrate the practical application of data structures in solving real-life problems, particularly in traffic management systems, and to show how prioritization can improve efficiency and response time in critical situations.

Problem Statement:

In real-world traffic systems, vehicles arriving at a signal are generally managed in a sequential order based on the First-In-First-Out (FIFO) principle. However, this conventional approach does not consider the urgency of certain vehicles such as ambulances, fire trucks, and police vehicles, which require immediate passage to handle emergency situations.

The delay in allowing such emergency vehicles can lead to critical consequences, including loss of life and increased damage during emergencies. Therefore, there is a need for an efficient system that can manage normal traffic flow while also providing priority to emergency vehicles.

This project aims to design and implement a **Traffic Signal Simulation System using the Queue Data Structure in C programming**, where two separate queues are maintained—one for normal vehicles and another for priority vehicles. The system ensures that emergency vehicles are given higher priority and are allowed to pass the signal before normal vehicles, thereby improving the efficiency and realism of traffic management simulation.

Objectives:

The main objectives of this project are:

- To simulate a real-world traffic signal system using the Queue data structure in C programming.
- To implement the **First-In-First-Out (FIFO)** principle for managing vehicle flow at traffic signals.
- To introduce a **priority-based mechanism** for handling emergency vehicles such as ambulances, fire trucks, and police vehicles.
- To maintain two separate queues: a **priority queue** for emergency vehicles and a **normal queue** for regular vehicles.
- To ensure that emergency vehicles are given immediate access before normal vehicles during signal clearance.
- To demonstrate the practical application of data structures in solving real-life traffic management problems.
- To improve understanding of queue operations such as insertion (enqueue), deletion (dequeue), and traversal through simulation.

Data Structure Used:

The primary data structure used in this project is the **Queue**.

Queue (FIFO – First In First Out)

A queue is a linear data structure in which insertion of elements takes place at the **rear end**, and deletion of elements takes place from the **front end**. It follows the **First-In-First-Out (FIFO)** principle, meaning the vehicle that arrives first is served first.

In this project, the queue is used to simulate vehicles waiting at a traffic signal.

<u>Types of Queues Used in This Project</u>

To implement priority handling, two separate queues are used:

1. Priority Queue

- Used for emergency vehicles such as:
  - Ambulance
  - Fire Truck
  - Police Vehicle
- These vehicles are given higher priority and are processed first when the signal turns green.

2. Normal Queue

- Used for regular vehicles such as:
  - Cars
  - Bikes
  - Trucks

- These vehicles follow normal FIFO order and are processed after priority vehicles.

<u>Operations Performed on Queue</u>

- **Enqueue (Insertion):** Adding a vehicle to the queue
- **Dequeue (Deletion):** Removing a vehicle when it passes the signal
- **Display:** Showing all waiting vehicles in the queue

Algorithm:

The algorithm for the **Traffic Signal Simulation with Priority Vehicles** is as follows:

Step 1: Start

Begin the execution of the program.

Step 2: Initialize Queues

- Create two queues:
  - **Priority Queue** (for ambulance, fire truck, police)
  - **Normal Queue** (for cars, bikes, trucks)
- Initialize front and rear of both queues to -1.

Step 3: Vehicle Arrival

- Accept vehicle name/type as input.
- Check the type of vehicle:
  - If it is an **emergency vehicle**, insert it into the **Priority Queue**.
  - Otherwise, insert it into the **Normal Queue**.

Step 4: Enqueue Operation

- If queue is not full:
  - Add vehicle at the rear position.
  - Update rear pointer.
- If queue is empty initially:
  - Set front = 0 and rear = 0.

Step 5: Signal Turns GREEN

- When signal becomes GREEN:
  - Check **Priority Queue first**.
    - If not empty → remove vehicle from Priority Queue (Dequeue).
  - Else check **Normal Queue**.

- If not empty → remove vehicle from Normal Queue (Dequeue).

Step 6: Dequeue Operation

- Remove the vehicle from the front of the queue.
- Update front pointer.
- If queue becomes empty, reset front and rear to -1.

Step 7: Display Queue

- Traverse both queues from front to rear.
- Display all waiting vehicles.

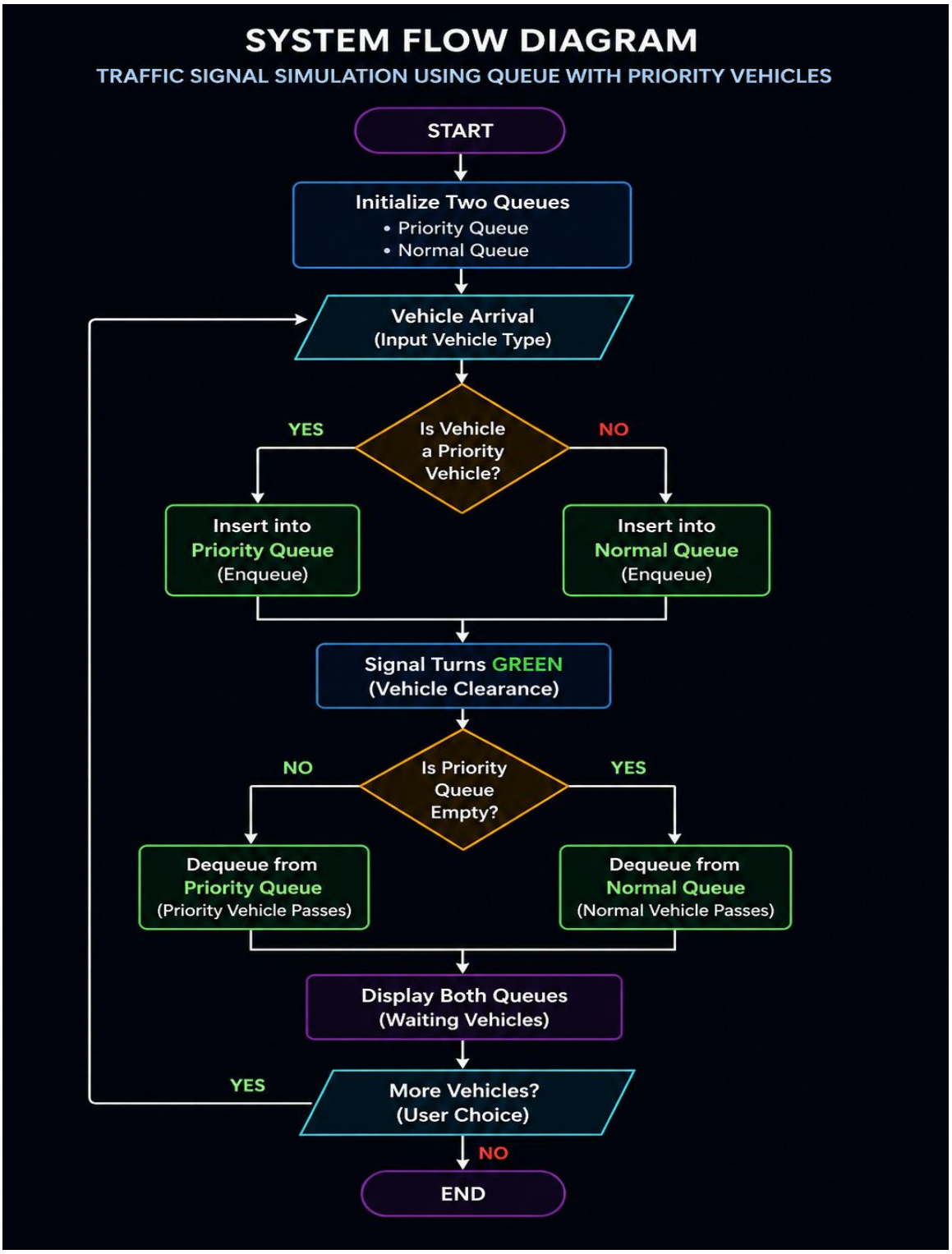
Step 8: Repeat Process

- Continue the process for multiple vehicle arrivals and signal cycles until the user exits.

Step 9: Stop

End the program execution.

System WorkFlow:



Implementation:

Normal Traffic Cycle

- The traffic signal follows a fixed sequence: North → East → South → West.
- Each direction is given a green signal for a fixed duration while all other directions remain red.
- This cycle simulates standard traffic light operation in real-world intersections.

Emergency Vehicle Handling

- When an emergency vehicle (such as an ambulance or fire truck) is detected, the system immediately overrides the normal cycle.
- The corresponding direction is given a green signal while all other directions are turned red.
- After the emergency vehicle passes, the system resumes normal traffic flow.

Control Mechanism

- A flag variable is used to indicate emergency mode.
- Based on this flag, the system dynamically switches between normal operation and priority mode.

Time Simulation

- The sleep() function is used to simulate real-time delays between signal changes, making the system behave closer to real traffic lights.

Advantages and Limitations:

Advantages:-

1. **Simple and Efficient Simulation**
   - The system effectively simulates real-world traffic signal behavior using a simple program structure.
2. **Emergency Vehicle Priority**
   - Provides priority handling for emergency vehicles like ambulances and fire trucks, improving realism and usefulness.
3. **Easy to Understand and Implement**
   - Uses basic C concepts such as loops, functions, and conditional statements, making it beginner-friendly.
4. **Low Resource Consumption**
   - The program is lightweight and does not require any advanced hardware or software resources.
5. **Scalable Design**
   - The logic can be extended easily to include more advanced features like sensors or real-time data input.

Limitations:-

1. **No Real Hardware Integration**
   - The system is purely software-based and does not interact with actual traffic lights or sensors.
2. **Fixed Timing System**
   - Signal timings are static and do not adapt to real-time traffic density.
3. **No Graphical Interface**

- The output is console-based, which makes visualization less interactive.

4. **Limited Intelligence**
    - The system does not use AI or advanced decision-making algorithms for traffic optimization.
5. **Manual Emergency Input**
    - Emergency vehicle detection is simulated manually rather than using automated detection systems.

Conclusion:

The Traffic Signal Simulation project successfully demonstrates the working of a basic traffic management system using the C programming language. It simulates real-world traffic control by following a fixed signal cycle for multiple directions and also incorporates a priority mechanism for emergency vehicles such as ambulances and fire trucks.

The project helps in understanding how fundamental programming concepts like loops, conditional statements, functions, and control variables can be applied to solve real-life problems. It also highlights the importance of emergency handling in traffic systems, ensuring faster clearance for critical vehicles.

Although the current implementation is simple and console-based, it provides a strong foundation for developing more advanced and intelligent traffic control systems in the future.